

Lecture 2. Euclidean quantum fields on $\mathbb{R}^d$

Last time: Gaussian field (OU) on $\mathbb{R}$

① View Euclidean time as "space x"

X is a spatial field. with correlation

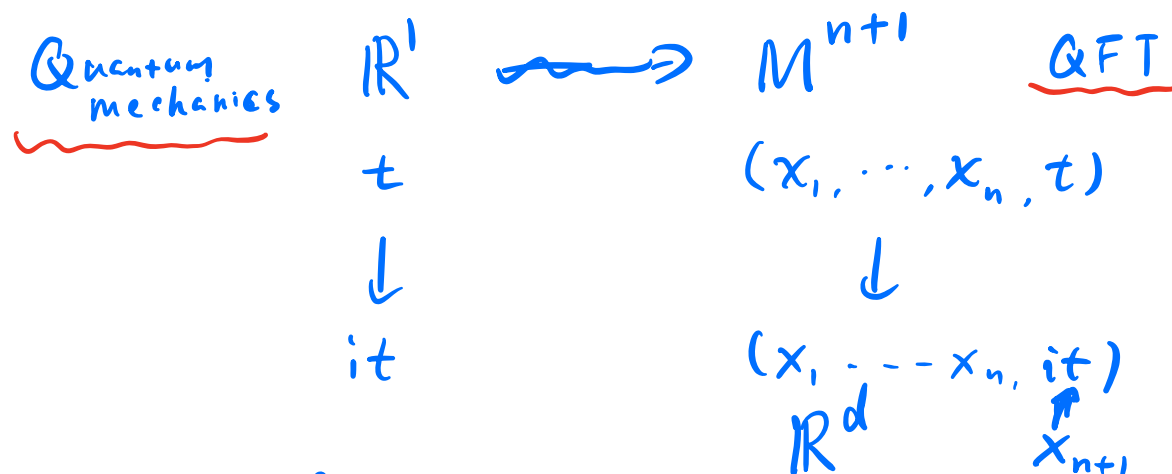
$$\frac{e^{-\alpha |x_1 - x_2|}}{2\alpha} \xrightarrow{\text{Fourier}} \frac{1}{\alpha^2 + k^2}$$

So one can formally write this Gaussian measure as on $\{\phi\}$

$$e^{-\frac{1}{2} \int |\hat{\phi}(k)|^2 (\alpha^2 + k^2) dk} = e^{-\frac{1}{2} \int (\alpha \phi)^2 + \alpha^2 \phi^2 dx}$$

properties of Fourier
e.g. $\partial_x \rightarrow ik$
Parseval identity

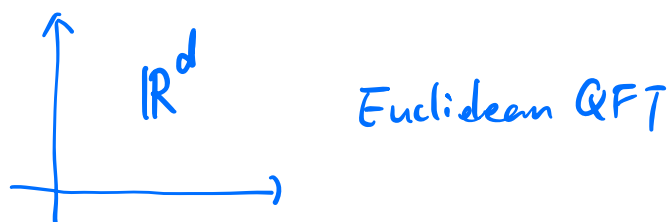
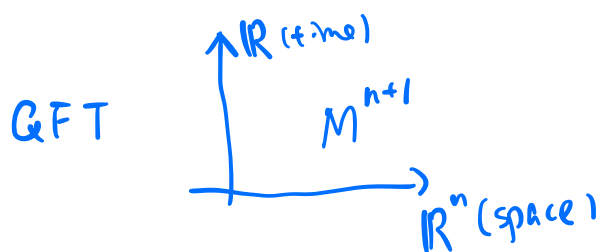
② Above formula obviously generalize to high dim.
 ~~~~~> Quantum field theory (QFT) (see next page)



Mink-metric  $x_1^2 + \dots + x_n^2 - t^2$

$= x_1^2 + \dots + x_n^2 + t^2 = \text{Euclidean metric on } \mathbb{R}^d$

$d = n+1$



Definition:

GFF with  $m > 0$  on  $\mathbb{R}^d$  (as an Euclidean QFT)

is a Gaussian measure  $N(0, C)$  on  $\mathcal{D}'(\mathbb{R}^d)$ ,

$$\text{where } (FC)(k) = \frac{1}{m^2 + k^2}$$

Namely, it's a random field  $\Phi$  s.t  $\forall f, g \in \mathcal{D}(\mathbb{R}^d)$

$$\mathbb{E}[\Phi(f)] = 0$$

$$\mathbb{E}[\Phi(f) \Phi(g)] = \int \frac{\hat{f}(k) \hat{g}(k)}{k^2 + m^2} dk$$

$$\mathbb{E}\left[\int \hat{\Phi}(k) \hat{f}(k) dk \int \hat{\Phi}(k') \hat{g}(k') dk'\right]$$

Existence of GFF ( $m > 0$ ) can be shown in a few ways:

e.g. 1) directly on  $\mathbb{R}^d$  2) approx by lattice GFF

1) Characteristic function (Fourier) for measure  $\mu$  on  $\mathcal{D}'(\mathbb{R}^d)$

$$S(f) = \int_{\mathcal{D}'(\mathbb{R}^d)} e^{i\Phi(f)} d\mu(\Phi) = \mathbb{E} e^{i\Phi(f)} \quad \forall \Phi \in \mathcal{D}(\mathbb{R}^d)$$

properties:

$$(1) S(0) = 1$$

(2)  $S(f)$  is continuous in  $f$

$$(3) \text{positive def: } \sum_{i,j=1}^n z_i \bar{z}_j \mathbb{E}(\Phi(f_i) - \Phi(f_j)) \geq 0$$

$$\forall f_i \in \mathcal{D}(\mathbb{R}^d) \\ z_i \in \mathbb{C}$$

Conversely:

Bochner-Minlos thm If a function  $S$  on  $(\mathbb{R}^d)$  satisfies (1)(2)(3)

Then  $\Rightarrow$  Borel Prob. measure  $\mu$  on  $\mathcal{D}'(\mathbb{R}^d)$

$$\text{st } S(f) = \int e^{i\varphi(f)} d\mu(\varphi)$$

pf: skip. [Glimm-Jaffe P54] or [Simon 79. Thm 2.3]

Essentially, use finite-dim thm (Bochner) + Kolmogorov extension.

For GFF,  $S(f) = e^{-\frac{1}{2}(f, Cf)_{L^2(\mathbb{R}^d)}}$

This uniquely determines a measure  $\mu$ :

Check (3):  $f_1, \dots, f_n \in \mathcal{D}(\mathbb{R}^d)$

Choose ON basis  $e_1, \dots, e_m$  i.e.  $(e_i, Ce_j)_{L^2} = \delta_{ij}$   
of  $\text{span}(f_1, \dots, f_n)$

$$Z = (Z_1, \dots, Z_m) \text{ indep } N(0,1)$$

Then  $\phi(\alpha_1 e_1 + \dots + \alpha_m e_m) = e^{-\frac{1}{2} \sum_{i=1}^m \alpha_i^2} = \mathbb{E} e^{i \vec{\alpha} \cdot \vec{Z}}$   
 $\forall \vec{\alpha} \in \mathbb{R}^m$

$$\sum_{i,j=1}^n z_i \bar{z}_j \underbrace{\phi(f_i - f_j)}_{\mathbb{E}[e^{i(f_i - f_j) \cdot Z}]} = \mathbb{E} \left| \sum_{j=1}^n z_j e^{i f_j \cdot Z} \right|^2 \geq 0$$

2) Approximate by Lattice GFF.

Let  $(X^{\varepsilon, M}_{(x)})_{x \in \Lambda_{\varepsilon, M}}$  be Gaussian on  $\mathbb{R}^{|\Lambda_{\varepsilon, M}|} = \mathbb{R}^{(\frac{M}{\varepsilon} + 1)^d}$

$$\begin{cases} \mathbb{E}[X^{\varepsilon, m}(x)] = 0 \\ \mathbb{E}[X^{\varepsilon, m}(x) X^{\varepsilon, m}(y)] = (m^2 - \Delta_\varepsilon)^{-1}(x, y) \end{cases}$$

Denote law of  $X^{\varepsilon, m}$  by  $\mu^{\varepsilon, m}$ .

$$\Delta_\varepsilon f(x) = \varepsilon^{-2} \sum_{i=1}^d f(x + \varepsilon e_i) - 2f(x) + f(x - \varepsilon e_i)$$

$$(m^2 - \Delta_\varepsilon)^{-1}(x, y) = \frac{1}{M^d} \sum_{k \in \hat{\Lambda}_{\varepsilon, m}} \frac{e^{2\pi i k \cdot (x-y)}}{m^2 + \sum_i (2\varepsilon^{-1} \sin(\pi \varepsilon k_i))^2}$$

$$\nabla_\varepsilon^i f(x) = \frac{1}{\varepsilon} (f(x + \varepsilon e_i) - f(x))$$

$$\nabla_\varepsilon^{-i} f(x) = \frac{1}{\varepsilon} (f(x) - f(x - \varepsilon e_i))$$

$$(\nabla_\varepsilon^i)^* = -\nabla_\varepsilon^{-i} \quad \Delta_\varepsilon = \sum_{i=1}^d \nabla_\varepsilon^{-i} \nabla_\varepsilon^i$$

$$(\nabla_\varepsilon^i \mathcal{F}^{-1} g)(x) = \int_{\hat{\Lambda}_\varepsilon} g(k) \frac{1}{\varepsilon} (e^{2\pi i \varepsilon k_i} - 1) e^{2\pi i k \cdot x} dk$$

$$(\nabla_\varepsilon^{-i} \nabla_\varepsilon^i \mathcal{F}^{-1} g)(x) = \int_{\hat{\Lambda}_\varepsilon} g(k) \frac{(2\varepsilon^{-1} \sin(\pi \varepsilon k_i))^2}{\approx (2\pi k_i)^2} e^{2\pi i k \cdot x} dk$$

exercise or  
TA session

One can show  $\mu^{\varepsilon, m} \rightarrow \mu$  on  $\mathcal{D}'(\mathbb{R}^d)$

(we skip this, since we'll do the harder  $\mathbb{R}^4$  later.)

But, several important remarks:

① Recall quantum mechanics:

$$\mathbb{R}^1 \xrightarrow{t \rightarrow -it} \mathbb{R}^1$$

quantum harmonic oscillator  $\rightarrow$  OU

Now QFT:

$$\mathbb{R} \times \mathbb{R}^n \xrightarrow{t \rightarrow -it} \mathbb{R}^d$$

infinitely many quantum harmonic oscillators  $\rightarrow$  many OUs

Recall single OU<sub>d</sub> :  $\frac{e^{-\alpha|t|}}{2\alpha} \xrightarrow{\text{Fourier}} \frac{1}{\alpha^2 + \omega^2}$

For each Fourier  $p \in \mathbb{R}^n$ , we have a complex OU. i.e.

$$X(t, x) = \int_{\mathbb{R}^n} \hat{X}(t, p) e^{ix \cdot p} dp$$

$$\text{where } \begin{cases} \hat{X}(t, p) = \frac{1}{\sqrt{2}} (Y_1(t, p) + i Y_2(t, p)) \\ \hat{X}(t, -p) = \frac{1}{\sqrt{2}} (Y_1(t, p) - i Y_2(t, p)) \end{cases}$$

$Y_1(p), Y_2(p)$   
indep OU

with  
 $\alpha_p^2 = p^2 + m^2$   
(see KG equ below)

$$\text{So, } \mathbb{E}[\hat{X}(t, p) \hat{X}(t', p')]$$

$$\begin{aligned} &= \frac{1}{2} \delta_{p+p'=0} \mathbb{E}[Y_1(t, p) Y_1(t', p) + Y_2(t, p) Y_2(t', p)] \\ &= \delta_{p+p'=0} \frac{e^{-\alpha_p |t-t'|}}{2\alpha_p} \end{aligned}$$

$$\mathbb{E}[X(t, x) X(t', x')] = \iint_{\mathbb{R}^n \times \mathbb{R}^n} \mathbb{E}[\hat{X}(t, p) \hat{X}(t', p')] e^{ix \cdot p + ix' \cdot p'} dp dp'$$

$$= \int_{\mathbb{R}^n} \frac{e^{-\alpha_p |t-t'|}}{2\alpha_p} e^{i(x-x') \cdot p} dp$$

$$= \int_{\mathbb{R}^n} \int_{\mathbb{R}} \frac{1}{\underbrace{p^2 + \omega^2 + m^2}_{k^2}} e^{i(t-t')\omega} e^{i(x-x') \cdot p} d\omega dp$$

where  $k = (\omega, p) \in \mathbb{R}^{n+1} = \mathbb{R}^d$ .

So  $X$  is GFF on  $\mathbb{R}^d$ .

This naturally arise in "quantizing" Klein-Gordonequation (relativistic)  
 $(\partial_t^2 - \Delta + m^2) u = 0$

② If  $m=0$ ,  $d=1, 2$ ,  $\int_{[1, \infty]^d} \frac{1}{k^2 + m^2} dk = \infty$   
as  $M \rightarrow \infty$

Massless GFF has no infinite volume limit in  $d=1, 2$ .

(see Friedli, Velenik's book) Infrared problem

(This is related with recurrence of RW in  $d=1,2$ )

(But, "modulo a constant".

it can be defined in infinite volume, see Berestycki-Powell)

Exercise: Brownian Bridge

as  $T \rightarrow \infty$  Variance at 0 blows up

But it remains finite as  $T \rightarrow \infty$  for our bridge.

In this course, we always assume  $m > 0$ .

③ GFF  $u$  on  $\mathcal{D}'(\mathbb{R}^d)$

In  $d=1$ .  $u$  support on  $C(\mathbb{R})$

Ultraviolet problem

But if  $d \geq 2$ .  $u$  does not supp on "functions"

$$\xrightarrow{\varepsilon \rightarrow 0} (m^2 - \Delta_\varepsilon)^{-1}(x, x) = \frac{1}{m^d} \sum_{k \in (m^{-1}\mathbb{Z})^d} \frac{1}{m^2 + k^2} = \infty \quad \text{if } d \geq 2$$

So, GFF doesn't have pointwise values.

If time permits, we can show more examples of QFT.

$$e^{-\int (\nabla\phi)^2 + m^2\phi^2 + \phi^4 dx} \quad D\phi \quad \phi^4\text{-model.}$$

symmetry  $\phi \rightarrow -\phi$

$$e^{-\int (\nabla\phi)^2 + e^{\gamma\phi} dx} \quad D\phi$$

"conformal"

Hegh-Krohn model  
(Liouville model)

$$e^{-\int (\nabla\phi)^2 + \cos(\beta\phi)} \quad D\phi$$

$\phi \rightarrow \phi + \frac{2\pi k}{\beta}$

sine-Gordon model

$$e^{-\int \sum_{i=1}^N (\nabla\phi_i)^2 + m^2\phi_i^2 + \frac{1}{N} \left( \sum_{i=1}^N \phi_i^2 \right)^2 dx} \quad D\phi$$

rotation in  $\mathbb{R}^N$

linear  $\sigma$ -model

$$e^{-\int_{\mathbb{R}^2} |dA|^2 dx} \quad DA$$

$A = (A_1, A_2) \quad dA = \text{curl } A$

Abelian gauge theory

$$A \rightarrow A + \nabla f \quad \text{gauge symmetry}$$

$$e^{-\int_{\mathbb{R}^2} |dA|^2 + |D_A\phi|^2 + m^2\phi^2 + \phi^4 dx} \quad D\phi \quad DA \rightarrow \mathbb{C}\text{-valued}$$

$$A \rightarrow A + \nabla f$$

$$\phi \rightarrow e^{if} \phi$$

$$D_A\phi = \partial\phi + iA\phi$$

Abelian Higgs model.

These all suffer from same challenges.